BEAUTY & WELLNESS

Customer Sentiment Intelligence and Personalised Product Recommendations

A Data Science Business Problem Definition, Research Rationale and Proposed Analysis Plan

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Executive Summary

Beauty and wellness shoppers face a crowded market of skincare, haircare, cosmetics and personal-care products. Although online stores contain large volumes of customer reviews and ratings, businesses may struggle to convert this unstructured feedback into practical decisions. Average star ratings can hide the reasons behind satisfaction or dissatisfaction: a product may be praised for texture but criticised for fragrance, packaging, price, or suitability for a particular skin type. At the same time, generic recommendations can feel irrelevant and reduce confidence in the shopping experience.

This report proposes a data science project that combines customer-review sentiment analysis with explainable, preference-aware product recommendations. The aim is to identify the product attributes most associated with positive and negative experiences, detect recurring customer pain points, and design a prototype recommendation approach that uses product attributes and review insights. The work is planned around publicly available, appropriately licensed review data, with a small pilot followed by more robust evaluation. It is a research plan, not a claim that an analysis has already been executed; no results or model accuracy figures are invented in this document.

The proposed workflow includes data sourcing and documentation, data-quality checks, exploratory analysis, natural-language processing, sentiment and aspect-level analysis, recommendation baselines, model evaluation, ethical review, and business recommendations. Success will be measured through both technical metrics and business-oriented indicators, such as the usefulness of recommendations, the clarity of explanations, and the ability to identify actionable product-improvement themes.

1. Business Context and Industry Relevance

The beauty and wellness sector is highly experience-driven. Product outcomes and perceptions can vary by individual, expectations, usage habits, price sensitivity and product characteristics. Customers often use reviews to assess whether a product is worth trying, while brands use feedback to understand quality concerns, product-market fit and emerging preferences. Review data therefore represents a useful source of customer insight, provided it is analysed carefully and not treated as a perfect representation of every customer.

The data science opportunity is to move beyond a single average rating. Natural language processing (NLP) can classify the tone of reviews and identify the specific aspects being discussed. Recommendation methods can then use product metadata, declared preferences and aggregated review themes to produce more relevant suggestions. The intended outcome is decision support for a retailer or beauty brand—not medical advice or a guarantee that a product will work for an individual.

2. Selected Business Problem

Problem statement

A beauty e-commerce business may be unable to consistently identify the main drivers of customer satisfaction and dissatisfaction across product categories, and may provide generic product recommendations that do not adequately reflect customers' stated preferences. This can make it harder to prioritise product improvements, communicate product attributes clearly, and help shoppers compare suitable alternatives.

Proposed data science solution

Develop a prototype analytical framework that (1) classifies customer review sentiment, (2) extracts sentiment about relevant product aspects such as texture, fragrance, packaging, value, perceived effectiveness and suitability, and (3) ranks candidate products using transparent preference and product-attribute signals. The prototype should provide interpretable outputs, for example: “frequently praised for lightweight texture” or “recurring complaints about packaging,” with adequate sample-size safeguards.

Why this problem is appropriate

- It connects a recognisable business need with clearly defined data science methods.
- Customer reviews and product metadata can be studied through public datasets, subject to licence and usage conditions.
- The project can start with simple, explainable baselines and expand only if data quality and time permit.
- It supports both marketing decisions and customer-experience improvements.
- It is measurable through classification metrics, recommendation metrics, qualitative review of errors and stakeholder-oriented usefulness checks.

3. Objectives and Scope

Primary objective

Design and evaluate a reproducible, interpretable analysis plan for turning beauty-product reviews into actionable customer insights and more relevant product recommendations.

Specific objectives

1. Describe the review dataset, product categories, ratings, review volume and data-quality limitations.
2. Measure overall sentiment and, where labels or annotation permit, sentiment at product-aspect level.
3. Identify common positive and negative themes by product category, brand or price band when these fields are available and sufficiently populated.
4. Build a baseline sentiment classifier and compare it with a stronger NLP model only if resources allow.
5. Develop a transparent recommendation baseline using product attributes and/or user–item interaction history, depending on the available fields.
6. Evaluate model quality, error patterns, potential bias, privacy risks and business usefulness.
7. Translate findings into prioritised recommendations for product teams, customer support and marketing.

Scope boundaries

The initial project focuses on online beauty and skincare product reviews. It does not attempt to diagnose skin conditions, infer sensitive health attributes, or promise that a product is safe or effective for an individual. If the dataset does not include reliable skin-type, ingredient or user-preference fields, those attributes will not be assumed. Recommendation explanations will be limited to evidence available in the data.

4. Research Questions and Hypotheses

ID	Research question	Proposed hypothesis / testable expectation
RQ1	Which product aspects appear most frequently in positive and negative reviews?	Aspect frequency and sentiment differ across product categories; test using aspect-level counts and sentiment proportions.
RQ2	How closely do star ratings align with text-based sentiment?	Ratings and text sentiment will often agree, but disagreement cases will reveal useful nuances.
RQ3	Can an NLP model classify review sentiment reliably on held-out data?	A trained model should outperform a simple majority-class baseline; performance must be demonstrated using a held-out test set.
RQ4	Can preference-aware ranking produce more relevant recommendations than a popularity-only baseline?	A content-based or hybrid method may improve ranking metrics for users/items with sufficient data.
RQ5	Do review themes differ by price band, brand or category?	Differences may exist, but conclusions should be reported only when group sizes and data coverage are adequate.

These are hypotheses to test, not findings. They should be refined after the dataset is inspected. Correlations between reviews, ratings and product attributes do not by themselves establish causation.

5. Data Sources and Data Requirements

The preferred approach is to select one primary beauty/skincare review dataset with clear provenance and a licence permitting the intended analysis. A secondary dataset may be used for a limited robustness check if time allows. Dataset documentation and licence terms should be reviewed before use.

Candidate public datasets

- **Aspect-Based Sentiment Analysis Dataset of Skincare Product Reviews (Mendeley Data, 2026).** A dataset described as containing 2,003 aspect-level annotated instances, with product/category, rating, date, review text and aspect-sentiment annotations. This is a practical option for a focused aspect-level pilot. *https://data.mendeley.com/datasets/6vmrcypy7c/1*
- **RecSys_Dataset: Beauty Product Reviews (Mendeley Data, 2025).** A synthetic dataset described as 50,000 reviews across 200 products and 1,000 users, with sentiment and interaction-related fields. Because it is synthetic, it is useful for prototyping but should not be presented as direct evidence of real market behaviour. *https://data.mendeley.com/datasets/n9sk45scmt/1*
- **Sephora Skincare Reviews (Innovatiana dataset page).** A large skincare-review resource described as containing review text and product metadata. Confirm access, provenance, licence, coverage and permitted use before selecting it. *https://www.innovatiana.com/en/datasets/sephora-skincare-reviews*

Required variables

Variable	Purpose	Required / optional
Review text	Sentiment and aspect/theme	Essential

	extraction	
Rating (e.g., 1–5)	Rating distribution; weak label or comparison target	Strongly preferred
Product ID/name and category	Grouping reviews and describing product types	Essential for product-level analysis
Aspect sentiment labels	Evaluate aspect-level model or build supervised pilot	Optional but valuable
Review date	Trend analysis over time	Optional
Brand and price	Segment comparisons and recommendation context	Optional
User ID / anonymised interaction history	Collaborative recommendation and user–item evaluation	Optional; only use where lawful and appropriate
Product attributes/ingredients	Content-based ranking and explanation	Optional; do not infer missing values
Helpful votes / verified-purchase flag	Sensitivity analysis and review-quality context	Optional; not a guarantee of truthfulness

6. Data Preparation and Quality Assurance

8. Record the source, download date, licence, fields, known collection method and any restrictions.
9. Inspect row count, column types, missingness, duplicate records, empty reviews, unusual ratings and inconsistent product identifiers.
10. Remove exact duplicates where justified; preserve an audit count and document every exclusion.
11. Normalise text carefully: handle HTML fragments, whitespace, encoding and obvious spam without deleting meaningful negation or product terminology.
12. Keep raw text separate from cleaned text so that transformations are reproducible.
13. Check class imbalance in sentiment labels and rating distributions. Do not silently convert neutral or mixed reviews into positive/negative labels.
14. Prevent data leakage: split at review level and, where appropriate, group by user or product so near-duplicate or same-user records do not cross train/test boundaries.
15. Use only the minimum necessary personal data; remove direct identifiers and avoid attempting to re-identify reviewers.

7. Exploratory Data Analysis Plan

The first analysis phase will answer what the dataset contains before any complex modelling is attempted.

- Dataset profile: total reviews, unique products, categories, date range and missing-value percentages.
- Rating profile: frequency of each rating and average/median rating by category, with review counts displayed.
- Text profile: review-length distribution, frequent terms and common phrases after careful cleaning.
- Sentiment profile: positive/neutral/negative proportions, overall and by category, where labels are available or generated with a validated method.
- Aspect profile: counts and sentiment by texture, fragrance, packaging, perceived effectiveness, value, suitability and other aspects supported by the corpus.

- Time trends: monthly review volume and sentiment only if dates are sufficiently complete and comparable.
- Segment comparisons: brand or price-band comparisons only when metadata quality and sample sizes justify them.

Suggested visualisations include a rating bar chart, sentiment distribution chart, missing-data summary, review-volume trend, aspect-by-sentiment heatmap and a table of recurring positive/negative themes. Charts should show denominators and sample sizes to avoid overinterpreting small groups.

8. Proposed Data Science Methods

8.1 Sentiment classification

Begin with a simple baseline such as majority-class prediction and a TF–IDF representation with Logistic Regression or Linear Support Vector Classification. If labels are reliable and computational resources allow, compare a pre-trained transformer model. Use stratified train/validation/test splits where appropriate, and keep the final test set untouched until model selection is complete. Report macro-F1, per-class precision and recall, confusion matrix and accuracy; accuracy alone may be misleading under class imbalance.

8.2 Aspect-based sentiment analysis

If aspect labels are supplied, evaluate an aspect classifier against those annotations. If not, create a small, documented manual annotation sample with a clear codebook and, if possible, a second reviewer to assess agreement. Candidate aspects may include texture, fragrance, packaging, value, perceived effectiveness and suitability, but the final categories must reflect the actual corpus. Report aspect-level precision/recall/F1 or clearly label exploratory keyword/theme extraction as unsupervised rather than validated classification.

8.3 Product recommendation

Start with a popularity or category-based baseline. Next, use content-based similarity from available product attributes and review-derived themes. If anonymised user–item interaction history is available and sufficiently dense, test collaborative filtering or a hybrid approach. For users with no history or new products, provide a cold-start fallback based on explicitly selected preferences and product metadata. Do not use sensitive attributes or infer skin conditions from review text.

8.4 Explainability

Each recommendation should provide a short evidence-based reason, such as a match to a selected product category, price range or frequently discussed attribute. Aggregate review themes only when a minimum sample threshold is met; show the number of reviews and avoid claiming that online reviews guarantee suitability.

9. Evaluation Framework

Component	Metrics / checks	Interpretation
Sentiment classifier	Macro-F1, per-class precision/recall, confusion matrix, accuracy	Shows whether the model recognises each sentiment class,

		including minority classes.
Aspect extraction/classification	Aspect-level F1; annotation agreement; error review	Tests whether the system identifies relevant product attributes and sentiment.
Recommendation ranking	Precision@K, Recall@K, NDCG@K or ranking relevance on a valid held-out interaction set	Measures whether relevant items appear near the top; only applicable when suitable interaction labels exist.
Business usefulness	Human review of insight clarity, actionable themes, recommendation explanations	Checks whether outputs could support real decisions; a prototype does not prove business impact.
Robustness/fairness	Performance by category and review volume; error analysis; cold-start behaviour	Identifies uneven performance and cases where the model should abstain.

Before analysis begins, establish baselines and document acceptance criteria with the intended stakeholder. Any numerical target should be treated as a project goal, not a promised outcome. A/B testing or a carefully designed user study would be needed to establish whether recommendations actually improve conversion or satisfaction.

10. Tools and Technical Stack

Tool	Role in the project
Python	Primary language for reproducible analysis.
Pandas / NumPy	Data loading, cleaning, feature preparation and descriptive statistics.
Matplotlib / Seaborn	Exploratory charts and quality checks.
scikit-learn	Train/test splitting, TF–IDF, baseline classifiers and evaluation.
NLTK or spaCy	Optional text processing and linguistic utilities.
Transformers (Hugging Face)	Optional pre-trained NLP model if justified by available compute and licence.
Jupyter Notebook / Google Colab	Exploration, experiment documentation and reproducibility.
Git / GitHub	Version control, README, code and non-sensitive project documentation.

11. Step-by-Step Work Plan (30–35 Hours)

Phase	Activities	Estimated time	Deliverable
1. Define & research	Confirm business context, review literature, select dataset and check licence.	4 h	Problem definition and source log
2. Data audit	Load data, inspect schema, missingness, duplicates, labels and privacy concerns.	4 h	Data dictionary and quality report
3. EDA	Summarise ratings, categories, text and sentiment; create initial charts.	5 h	EDA notebook and visualisations
4. NLP baseline	Build a simple sentiment baseline; compare models if labels permit.	6 h	Metrics, confusion matrix and error review

5. Aspect insights	Analyse product attributes using labelled data or a documented annotation sample.	4 h	Aspect-level insight table
6. Recommendation prototype	Create popularity/content-based baseline; add hybrid method only if data supports it.	5 h	Prototype ranking and explanations
7. Evaluation & ethics	Evaluate, inspect failure cases, assess bias, limitations and safe use.	3 h	Evaluation and risk section
8. Report & reproducibility	Write conclusions, recommendations, README and references.	3 h	Final report and project files
Total	**Planned workload; adapt based on dataset quality and actual progress.**	**34 h**	**Documented project package**

12. Expected Deliverables

- A well-documented dataset selection rationale and data dictionary.
- A reproducible exploratory analysis notebook with clearly labelled charts.
- A baseline sentiment model and evaluation report, if labelled data supports supervised learning.
- A product-aspect insight summary with sample sizes and limitations.
- A recommendation prototype or, if interaction data is inadequate, a documented content-based ranking demonstration.
- A short list of business recommendations ranked by expected usefulness and feasibility.
- A README describing setup, data source, licence, methods, limitations and how to reproduce the work.

13. Potential Business Value and Recommendations

If the analysis supports the expected patterns, the business could use the findings to:

- Prioritise recurring product complaints for investigation by product, packaging or customer-experience teams.
- Improve product pages by making frequently discussed attributes easier to compare.
- Create clearer, evidence-based product filters (for example, texture or fragrance preference) where reliable metadata exists.
- Provide customer support teams with aggregated emerging issue themes rather than relying on isolated anecdotes.
- Design targeted campaigns around verified product attributes and customer interests, without exaggerating product performance.
- Monitor changes in review themes over time, while accounting for product launches, seasonal effects and changes in review volume.

These are potential uses rather than guaranteed outcomes. Business impact should be validated with stakeholders and, where appropriate, controlled experiments or pilot testing.

14. Risks, Ethics and Limitations

Risk / limitation	Mitigation
Representativeness	Reviewers may not represent all buyers; review volume varies by product. Report coverage and avoid broad population claims.
Fake, incentivised or noisy reviews	Text and ratings may be manipulated or low quality. Use available provenance flags cautiously and run sensitivity checks.
Rating–text disagreement	A low rating may contain positive text or vice versa. Inspect disagreements instead of forcing one signal to dominate.
Bias and uneven coverage	Some categories, brands or user groups may be underrepresented. Report subgroup performance where feasible.
Privacy	Use only lawful, permitted data; minimise personal data and do not attempt re-identification.
Sensitive inference	Do not infer health conditions, ethnicity or other sensitive traits. Avoid clinical or dermatological recommendations.
Overclaiming	Sentiment is not proof of product efficacy, safety or causation. Clearly distinguish correlation, prediction and business impact.
Dataset licence and provenance	Confirm licence, terms, collection methods and attribution requirements before use; synthetic data should be labelled as synthetic.
Cold start	New products and users may lack interaction history. Provide a metadata-based fallback and explain uncertainty.
Language and context	Sarcasm, spelling variation, mixed languages and domain-specific terms can reduce model quality.

15. Research Foundation and Literature Review

Existing research supports the feasibility of analysing online reviews using NLP, sentiment classification and aspect-level methods. A 2024 survey by Malik and Bilal reviewed 154 publications on NLP for online customer reviews and described applications spanning sentiment analysis, review management, customer experience, user profiling and marketing/reputation management. Recent dataset releases also demonstrate that skincare reviews can be annotated at aspect level, enabling more granular evaluation than a single overall sentiment label. Research on sentiment-aware recommendation further motivates combining review text with product and interaction signals, while also highlighting the need for robust evaluation and attention to generalisability.

The proposed project is deliberately scoped as an interpretable, staged prototype rather than an attempt to reproduce a complex state-of-the-art architecture. Its practical contribution is the end-to-end framing: connect review themes to a defined business question, compare against simple baselines, evaluate recommendation relevance only where the data permits, and translate the outputs into cautious, actionable business recommendations.

16. Conclusion

Customer sentiment intelligence and personalised product discovery provide a clear, relevant data science problem for the beauty and wellness sector. By combining exploratory data analysis, sentiment and aspect-level analysis, and a transparent recommendation baseline, the project can help explain what customers discuss, where dissatisfaction may be concentrated, and how product choices might be presented more usefully. The approach is feasible within a 30–35-hour plan if the project begins with a focused dataset and simple baselines. The quality of conclusions will depend on data provenance, label reliability, coverage and evaluation design. The report therefore emphasises reproducibility, explicit limitations, privacy protection and evidence-based interpretation rather than claiming results before the analysis has been performed.

References and Public Resources

- Malik, H. and Bilal, M. (2024). Natural language processing for analyzing online customer reviews: a survey, taxonomy, and open research challenges. PeerJ Computer Science, 10, e2203. https://doi.org/10.7717/peerj-cs.2203
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- Ray, R. K. and Singh, A. (2025). From online reviews to smartwatch recommendation: An integrated aspect-based sentiment analysis framework. Journal of Retailing and Consumer Services, 82, 104059. https://doi.org/10.1016/j.jretconser.2024.104059

Appendix A. Submission Description (200+ Words)

This report proposes a data science project for the beauty and wellness industry focused on customer sentiment intelligence and personalised product recommendations. Online beauty shoppers rely heavily on product ratings and written reviews, but average ratings alone may not explain why customers like or dislike a product. Feedback can mention several distinct attributes, including texture, fragrance, packaging, perceived effectiveness, value for money and suitability. The proposed project will examine whether natural language processing and recommendation methods can convert these reviews into clear, useful business insights.

The work will begin by selecting a publicly available beauty or skincare review dataset and checking its provenance, licence, fields and limitations. The dataset will then be audited for missing values, duplicates, inconsistent ratings, class imbalance and privacy concerns. Exploratory data analysis will describe review volume, ratings, product categories and recurring terms. Where suitable labels are

available, a baseline sentiment classifier will be trained and evaluated using a held-out test set, macro-F1, per-class precision and recall, and a confusion matrix. Aspect-level analysis will be used to identify the product attributes most frequently associated with positive and negative feedback.

For product discovery, the project will compare a simple popularity baseline with a transparent content-based recommendation approach. Collaborative or hybrid filtering will only be attempted if appropriate anonymised interaction data is available. Evaluation will focus on technical performance, error analysis, recommendation relevance and the clarity of explanations. The final report will discuss potential business applications, including product-page improvements, prioritisation of recurring complaints and more relevant product comparisons. It will also address limitations such as biased reviews, incomplete metadata, cold-start problems and differences in language or context. The project will not infer medical conditions or claim that reviews prove product safety or efficacy. The expected deliverables are a documented analysis plan, reproducible notebook, evaluation summary and evidence-based business recommendations. No results will be claimed until the data has actually been analysed.

Appendix B. Suggested GitHub Repository Structure

```
beauty-sentiment-recommendation/
├── README.md
├── requirements.txt
├── data/
│   └── README_data_sources.md   # document source/licence; avoid uploading restricted data
├── notebooks/
│   ├── 01_data_audit_eda.ipynb
│   ├── 02_sentiment_baseline.ipynb
│   ├── 03_aspect_analysis.ipynb
│   └── 04_recommendation_baseline.ipynb
├── src/
│   ├── preprocessing.py
│   ├── sentiment.py
│   └── recommend.py
├── reports/
│   ├── figures/
│   └── evaluation_summary.md
└── LICENSE
```

Appendix C. Recommended README Checklist

- Project aim and business question
- Dataset source, licence, access date and attribution
- Environment setup and dependencies
- Steps to reproduce each notebook
- Train/test split strategy and random seeds
- Evaluation metrics and baseline comparison
- Known limitations and ethical safeguards
- Contact or author details, if appropriate